



**Dept. of Computer Science &
Engineering(AI&ML) and Computer Science
& Engineering (Cyber Security)**

**Programme : B E – Computer Science & Engineering(AI&ML) and Computer Science &
Engineering(Cyber Security)
Internal Assessment – 1**

TERM : 03/03/2025 to 21/06/2025	COURSE NAME: Automata Theory and Compiler Design
DATE : 25/04/2025 TIME: 3.00pm-4.00pm	COURSE CODE : CI44/CY44
MAX MARKS: 30	PORTIONS : L1-L20



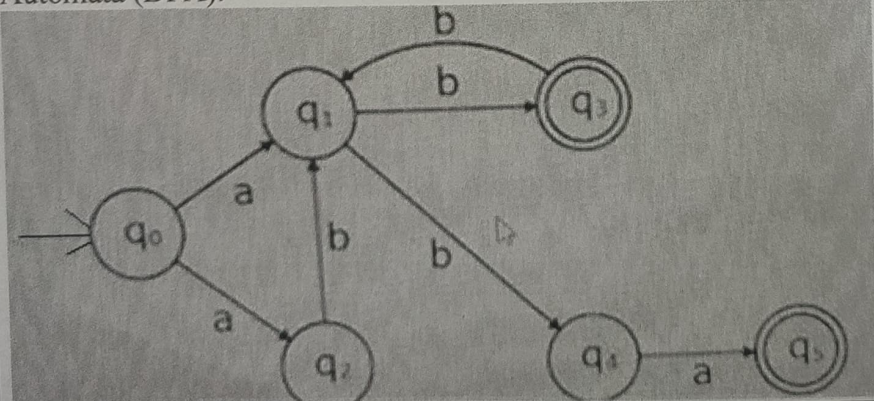
Mobile Phones are banned

Instructions to Candidates: Answer any TWO full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Compute FIRST of all Non-terminals for the grammar G $S \rightarrow aSe \mid B$ $B \rightarrow bBe \mid C$ $C \rightarrow cCe \mid d$	L3	2	3
b	I. Describe the languages denoted by the following regular expressions i. $(a b)^*a?(a b)$ ii. $a^*b^+a^*b^+a^+$ II. Write regular expressions using character class i. Accepting Email id for gmail/yahoo. ii. Accepting the DOB in the format (dd-mmm-yyyy).	L4	1	4
c	Construct Deterministic Finite Automata (DFA) for i. Strings consisting of a's and b's, which has exactly three a's. ii. Strings consisting of a's and b's, where each a is followed by atleast one b.	L3	1	4
d	Obtain Pushdown automata(PDA) to accept the language for $L = \{a^n b^{2n} \mid n \geq 1\}$	L3	3	4
2.a	i. For the expression grammar $E \rightarrow E * F \mid F + E \mid F$ $F \rightarrow F - \mid id$ The statement, which holds true, is _____ (A) + and - have same precedence (B) Precedence of * is higher + (C) Precedence of - is higher * (D) Precedence of + is higher * ii. Obtain the context free grammar to generate string consisting of any number of a's and b's with at least one b. iii. For the CFG given, check whether the given G is ambiguous for input "abd" G: $S \rightarrow AB \mid CD$ $A \rightarrow aAb \mid \epsilon$ $B \rightarrow dB \mid \epsilon$ $C \rightarrow aC \mid \epsilon$ $D \rightarrow bDd \mid \epsilon$	L4	3	1+2 +2 =5
b	Explain the phases of a compiler by translating the given assignment statement. (Assume all variables are of type int) $val = a * a + b * b;$	L2	1	5



c	Discuss why elimination of left recursion and left factoring are required for top down parser? Eliminate the left recursion from the following grammars. i. $S \rightarrow A \mid B$ $A \rightarrow Aa \mid \epsilon$ $B \rightarrow Bb \mid Sc \mid \epsilon$ ii. $S \rightarrow AB \mid Bb \mid c$ $A \rightarrow Bd \mid \epsilon$ $B \rightarrow d \mid e$	L3	2	1+2 +2 =5
3.a	Consider the Context Free Grammar given G: $A \rightarrow Bd \mid a$ $B \rightarrow Ae \mid b \mid \epsilon$ $C \rightarrow B \mid c$ Obtain the following for the input "ded" i. Rightmost derivation ii. Parse tree	L3	3	4
b	Compute FIRST of all Non-terminals for the grammars given i. $S \rightarrow aAS \mid \epsilon$ $A \rightarrow ba \mid SB$ $B \rightarrow ca \mid S$ ii. $S \rightarrow ABCD$ $A \rightarrow Bd \mid \epsilon$ $B \rightarrow b \mid \epsilon$ $C \rightarrow aBc \mid \epsilon$ $D \rightarrow d$ iii. $S \rightarrow Bc \mid DB$ $B \rightarrow ab \mid cS$ $D \rightarrow d \mid \epsilon$	L3	2	6
c	Convert the given Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA). 	L3	1	5